SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SIMATS ENGINEERING

Concept Mapping

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)

Max Marks:100

Annexure A

Concept Mapping

Code of Conduct:

I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Concept Mapping

1. Construct a comprehensive and well-organized concept map that analyzes the fundamentals of Computer Vision and Image Processing, including the overview, features, objectives, major components, and relationship between Computer Vision and Image Processing. Clearly represent the different levels of processing, distinguish low-level, mid-level, and high-level vision, and illustrate how information progresses from image input to meaningful interpretation. Establish meaningful relationships among the concepts and present the map with appropriate literature support, logical hierarchy, and visual clarity. (BTL-4)

2. Develop a detailed concept map that analyzes the digital image formation and acquisition process, covering image sensing, sensing devices, illumination, image formation models, image acquisition, sampling, and quantization. Clearly establish the dependencies among these stages and relate sampling and quantization to spatial resolution, intensity resolution, image quality, and digital image representation. Incorporate relevant literature perspectives and present the concepts using clear cause–effect, dependency, and hierarchical relationships. (BTL-4)

3. Design an analytically structured concept map that establishes the relationship between Image Processing and Computer Vision techniques and their real-world applications. Include major techniques such as image enhancement, restoration, segmentation, feature extraction, object detection, and recognition, and connect them to applications such as medical imaging, surveillance, autonomous systems, agriculture, industrial inspection, and other intelligent vision systems. Demonstrate how processing techniques contribute to application-specific objectives, integrate relevant literature perspectives, and maintain logical organization and visual clarity. (BTL-4)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand